

About the Prediction Assumption Standard

What Is the Prediction Assumption Standard?

The Prediction Assumption Standard (PAS) is the fourth Parent Standard within PGA™ — Prediction Governance Architecture .

It governs the assumptions that must hold, remain sufficiently plausible, or remain within defined conditions for a prediction and its stated probability, confidence, feasibility, or expected outcome to remain meaningful.

PAS exists because predictions do not depend only on methodology and information.

They also depend on conditions that are treated as holding.

These conditions may concern:

future events; market stability; system continuity; human behavior; operational conditions; regulation; environmental conditions; model applicability; data continuity; organizational decisions; external dependencies.

PAS makes those conditions explicit and governable.

Its foundational principle is:

A hidden assumption cannot remain a governed assumption.

Why the Standard Exists

A prediction can appear methodologically rigorous while still depending on assumptions that are never clearly stated.

For example, a forecast may implicitly assume that:

a supplier remains operational; regulation does not materially change; customer behavior remains comparable; a system continues functioning normally; historical relationships continue into the future; no major external disruption occurs.

If these assumptions remain hidden, users may believe the prediction is stronger than it actually is.

PAS creates a governed structure for identifying and maintaining the assumptions on which the prediction depends.

What PAS Governs

PAS governs the lifecycle of material predictive assumptions.

This includes:

assumption identification; explicit representation; assumption scope; assumption conditions; assumption origin; assumption state; assumption dependencies; assumption chains; assumption sensitivity; assumption materiality; critical assumptions; assumption uncertainty; assumption support; assumption conflict; assumption change; assumption breach; reassessment triggers; historical assumption preservation.

The objective is not to eliminate assumptions.

The objective is to make them visible and governable.



Prediction Assumption ≠ Established Fact

A prediction assumption may be reasonable, well-supported, and highly plausible.

It is still not automatically an established fact.

For example:

"Interest rates remain below 5% during the next six months."

may be a reasonable predictive assumption.

It remains an assumption because the future condition has not yet been established.

PAS preserves this distinction:

Prediction Assumption $\neq$ Established Fact

Prediction Basis $\neq$ Prediction Assumptions

PBAS governs:

what information the prediction is based on.

PAS governs:

what must hold for the prediction to remain meaningful.

The two spaces interact but remain different.

For example:

A prediction may use current supplier financial information as part of Prediction Basis.

It may simultaneously assume that:

the supplier continues operating throughout the Prediction Horizon.

The first is information.

The second is a predictive assumption.

Explicit and Hidden Assumptions

Material assumptions should be made explicit where they can reasonably be identified.

A hidden assumption can become dangerous because users may rely on the prediction without knowing that a critical predictive condition exists.

PAS therefore requires material implicit assumptions to be surfaced where possible.

Implicit Assumption $\neq$ Governed Assumption

The goal is not to document every imaginable condition.

It is to identify assumptions that materially affect prediction meaning.

Assumption Dependencies Matter

Assumptions rarely exist in isolation.

One assumption may depend on another.

For example:

Supplier remains financially solvent

may support:

Supplier maintains production capacity

which may support:

Required component deliveries continue

which may support:

Production target remains feasible.

This creates an assumption chain.

PAS makes these dependencies governable so that failure of one assumption can be understood in relation to the others.

Not Every Assumption Matters Equally

Some assumptions may have little effect on the prediction.

Others may fundamentally determine whether the prediction remains meaningful.

PAS therefore distinguishes:

low-impact assumptions; material assumptions; critical assumptions.

An assumption may also be unlikely to fail while still being highly material if failure would fundamentally change the prediction.

Low Probability of Failure $\neq$ Low Assumption Materiality

Assumptions Can Change

A prediction may begin with assumptions that are reasonable at formation time.

Those assumptions may later weaken, change, or fail.

For example:

regulation changes; market conditions deteriorate; a supplier fails; system configuration changes; operating conditions change; new information contradicts an earlier assumption.

PAS governs the assumption state and the conditions that should trigger reassessment.

It does not itself decide whether the entire prediction must be revised.

That broader prediction-level decision belongs later to PMRS — Prediction Monitoring & Revision Standard .

Assumption Breach $\neq$ Prediction Error

An assumption may fail while the prediction still happens to be correct.

Likewise, all assumptions may appear to hold while the prediction still proves wrong.

Therefore:

Assumption Breach $\neq$ Prediction Error

PAS governs predictive conditions.

It does not determine prediction accuracy.

Historical Assumptions Must Remain Visible

Later events must not rewrite the assumptions that existed when the prediction was formed.

If an assumption later proves wrong, the historical record should still show that the prediction originally depended on it.

This distinction is essential for legitimate review.

Later Knowledge $\neq$ Historical Assumption State

Position Within PGA™

PAS occupies the fourth governance space in the PGA™ lifecycle:

Prediction Object & Purpose → Prediction Methodology → Prediction Basis → Prediction Assumptions → Prediction Formation → Probability & Uncertainty → Prediction Feasibility → Collective Prediction where applicable → Communication & Reliance → Monitoring & Revision → Outcome & Performance → Prediction Record & Continuity

PAS receives a governed Prediction Basis and establishes the assumptions required before a specific prediction may be formed.

The transition is:

Governed Prediction Basis → Governed Prediction Assumption Set → Prediction Formation

How PAS Is Operationalized

PAS is implemented through five Core Modules:

PAIM — Predictive Assumption Identification Module

Identifies the material assumptions supporting the prediction and makes them explicit.

ADMM — Assumption Dependency Mapping Module

Maps relationships among assumptions and their dependencies on predictive conditions, inputs, methodologies, and other assumptions.

ASNM — Assumption Sensitivity Module

Determines how materially the prediction may change when an assumption changes, weakens, or fails.

AMAM — Assumption Materiality Module

Determines which assumptions materially affect prediction meaning, uncertainty, probability, feasibility, methodology applicability, or reliance.

ACGM — Assumption Change Governance Module

Governs assumption-state change, breach, reassessment, change propagation, and historical preservation.

The internal progression is:

Predictive Assumption Identification → Assumption Dependency Mapping → Assumption Sensitivity → Assumption Materiality → Assumption Change Governance

What PAS Does Not Govern

PAS does not govern:

Prediction Object definition; Prediction Methodology design; Prediction Basis; evidence validity; Operational Reality determination; Prediction Formation; probability representation; uncertainty representation; Prediction Feasibility; collective prediction legitimacy; prediction-level revision decisions; predictive performance; accountability allocation; final prediction validity.

PAS governs assumptions.

It does not govern the entire Prediction Lifecycle.

Cross-Architecture Boundaries

PBAS — Prediction Basis Standard governs the informational foundation of the prediction.

PFOS — Prediction Formation Standard governs how methodology, Prediction Basis, assumptions, and applicable conditions become a specific prediction.

PPUS — Prediction Probability & Uncertainty Standard governs probability, confidence, uncertainty, ranges, and conditionality.

PFES — Prediction Feasibility Standard governs whether the predicted outcome is realistically achievable under stated conditions.

PMRS — Prediction Monitoring & Revision Standard governs prediction-level review, revision, supersession, and withdrawal.

ORA™ governs Operational Reality.

ADIT® governs evidence governance.

PAS may consume relevant outputs from these spaces without absorbing their responsibilities.

Why PAS Matters

PAS allows organizations to understand not only:

what a prediction says

but also:

what must remain true for that prediction to retain its meaning.

This becomes especially important in:

AI forecasting; financial analysis; strategic planning; supply-chain forecasting; industrial systems; insurance; healthcare;

public policy; market prediction; operational decision systems.

Without assumption governance, predictive systems can hide fragility behind apparently precise outputs.

PAS exposes that fragility before it becomes invisible reliance.

Use Case 1 — A Supply Forecast Assumes the Supplier Will Still Exist

A manufacturer predicts that production will remain stable over the next six months.

The Prediction Basis contains strong historical delivery data, production records, demand forecasts, and supplier performance metrics.

The forecast appears robust.

However, the model implicitly assumes that the primary component supplier remains financially solvent and operational throughout the Prediction Horizon.

That assumption is not documented.

Three months later, the supplier enters insolvency proceedings and production is disrupted.

The critical governance question is:

Can the forecast be considered fully governed when a condition capable of fundamentally changing the outcome was never identified as an assumption?

PAS requires the supplier-continuity assumption to become explicit and governable.

Its dependency, sensitivity, and materiality can then be assessed.

A hidden assumption cannot remain a governed assumption.

Use Case 2 — An AI Forecast Quietly Assumes That Yesterday's World Continues Tomorrow

An AI system predicts customer demand using several years of historical purchasing behavior.

The system performs well during stable market conditions.

A major regulatory change then alters pricing rules and customer incentives.

The model continues using the same historical relationships without explicitly representing the assumption that prior behavioral patterns remain applicable.

The critical governance question is:

Can the prediction remain methodologically meaningful when it depends on continuity between historical behavior and future behavior but that continuity assumption has never been surfaced or reassessed?

PAS identifies the continuity relationship as a material predictive assumption.

If the assumption weakens or fails, its change can become visible and later trigger broader prediction review under PMRS.

Historical Stability $\neq$ Future Stability

Canonical Boundary

PAS governs the identification, explicit representation, scope, dependency, sensitivity, materiality, state, change, breach, historical preservation, and lifecycle relevance of assumptions materially supporting a prediction. It does not govern Prediction Basis itself, evidence validity, Prediction Methodology design, Prediction Formation, probability representation, uncertainty determination, Prediction Feasibility, Operational Reality determination, prediction-level revision decisions, predictive performance, or final prediction validity.

PAS governs:

what must hold for the prediction to remain meaningful.

It does not determine:

whether those conditions will actually hold.

Canonical Closing Statement

A prediction does not depend only on what is known. It also depends on what is assumed.

PAS makes those assumptions visible before they become invisible conditions of reliance.

A hidden assumption cannot remain a governed assumption.

Governed Links

[About Prediction Assumption Standard \(PAS\)](#)

[Prediction Assumption Identification Module \(PAIM\)](#)

[Assumption Dependency Mapping Module \(ADMM\)](#)

[Assumption Sensitivity Module \(ASNM\)](#)

[Assumption Materiality Assessment Module \(AMAM\)](#)

[Assumption Change Governance Module \(ACGM\)](#)

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